

INTERNAL · CONFIDENTIAL

# Santander Business Banking

## Concept Prototype — Technical Report

*A front-end exploration of paperless, compliant-by-design service journeys*

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### Contents

- 1 Executive summary
- 2 Context and problem statement
- 3 Objectives, scope and non-goals
- 4 System architecture
- 5 Workflow catalogue
- 6 Regulatory control mapping
- 7 Engineering and quality approach
- 8 Accessibility
- 9 Limitations and assumptions
- 10 Indicative next steps
- References

## SECTION 1

## 1 Executive summary

Business banking retains a striking dependence on paper. Mandate variations, account closures, beneficiary onboarding and dispute handling still routinely require posted forms, wet-ink signatures and branch attendance — processes that impose days of latency, measurable operational cost, and inconsistent application of controls at exactly the points where regulatory rigour matters most.

This report describes a self-initiated concept prototype that reframes seventeen such service events as guided, digitally-signed, self-serve journeys. The distinguishing idea is that the controls each journey must satisfy — Strong Customer Authentication [3], Confirmation of Payee [4], sanctions and politically-exposed-person screening [8], cooling-off, and record-keeping [12] — are expressed inside the flow rather than appended through manual back-office process. Control thereby becomes an enabler of a better experience instead of the reason to retain paper.

The prototype is deliberately a front-end artefact. It holds no customer data, connects to no live system, and is not a release candidate. Its purpose is to make a service proposition tangible enough to test — with colleagues, with risk and compliance, and with business customers — before any commitment to build.

| Dimension          | Summary                                                                                       |
|--------------------|-----------------------------------------------------------------------------------------------|
| Workflows          | Seventeen end-to-end paperless journeys                                                       |
| Entity coverage    | Seven business types (sole trader, partnership, limited company, LLP, charity, club, society) |
| Platform           | Single-file React front end, compiled to one self-contained HTML bundle                       |
| Regulatory framing | PSD2/SCA, CoP, PSR 2017, MLR 2017, FCA DISP/BCOBS/SYSC, Consumer Duty, UK GDPR, MTD           |
| Scope              | Front-end concept only — no backend, no real data, not production-hardened                    |

## SECTION 2

## 2 Context and problem statement

The friction in business-banking servicing is not primarily a technology gap; it is a process gap. The underlying capabilities — digital identity, strong authentication, Confirmation of Payee and real-time payments — are mature and, in several cases, mandated. Yet the service events that sit on top of them are frequently still discharged on paper, for three connected reasons: the controls are treated as separate from the customer journey; the journeys are owned by different teams; and no single artefact makes the end-to-end experience concrete enough to challenge.

The cost of that gap is cumulative. Postal round-trips convert minutes of decision-making into days of elapsed time. Manual re-keying introduces error and rework. And discretionary, manually-applied controls are, by construction, harder to evidence consistently — a growing concern under the FCA Consumer Duty [1], which expects firms to demonstrate good outcomes rather than assert them.

**THE CORE QUESTION**

What would these service events look like if they were re-imagined as in-app, self-serve, compliant-by-design journeys — with each regulatory check built into the flow, applied consistently, and recorded automatically?

**SECTION 3**

## 3 Objectives, scope and non-goals

The prototype was built to answer that question in a form that can be reviewed and tested, not merely described. Its objectives are therefore evidential rather than technical:

- Demonstrate credible paperless equivalents for the most paper-heavy business-banking service events.
- Express, within each journey, the specific regulatory control it must satisfy — so reviewers can see where an obligation is met rather than infer it.
- Provide a single, clickable artefact for internal review and structured customer research.
- Establish a consistent design language and a measurable accessibility standard for the journeys.

**Non-goals (explicitly out of scope)**

Equally important is what the prototype is not, so that its evidence is read correctly:

- No backend, database, or integration with live banking systems.
- No real customer, account, or payment data — every figure and reference is illustrative and fictional.
- No production security hardening; this is a concept artefact, not a release candidate.
- No processing of real payments, identity documents, or regulatory submissions.

**SECTION 4**

## 4 System architecture

The prototype is implemented as a single React component compiled by Vite into one self-contained HTML bundle. This is an unusual choice for production software but a deliberate one for a concept artefact: it keeps the entire product readable top-to-bottom, removes build-and-deploy friction, and lets the whole experience be shared as a single file. The trade-off — that all state lives in one component — is acceptable precisely because the artefact is not intended to scale as a codebase.

| Architectural choice                        | Rationale                                                                   |
|---------------------------------------------|-----------------------------------------------------------------------------|
| Single-file React front end                 | Whole product legible in one place; no module navigation to review it       |
| Compiled to one HTML artefact               | Frictionless to deploy, host, or hand to a colleague                        |
| Two shells, shared state (mobile & desktop) | Demonstrates responsive parity without a second codebase                    |
| Workflow-overlay model                      | Each journey is a self-contained, multi-step wizard over the main app       |
| Entity system (7 business types)            | Accounts, mandate rules, labels and required documents reconfigure per type |

**Mandate and authorisation model**

Authorisation is modelled explicitly. Each account carries a signing rule — any-one, any-two, or all signatories — and every workflow resolves the strictest applicable rule across the accounts it touches. That resolved rule determines whether an instruction can proceed immediately, requires a co-signatory, or enters a cooling-off period. Encoding authorisation as data rather than as bespoke per-journey logic is what allows the same seventeen journeys to behave correctly across all seven entity types.

## SECTION 5

# 5 Workflow catalogue

The prototype implements seventeen end-to-end workflows. Each replaces an identifiable paper or branch-based process and surfaces the control that governs it.

| Workflow                              | What it replaces / does                                       | Primary control             |
|---------------------------------------|---------------------------------------------------------------|-----------------------------|
| <b>Account closure</b>                | Posted closure request; in-credit and cooling-off checks      | BCOBS [11]                  |
| <b>Business details update</b>        | Change of name/address/contact with proof of address          | KYB [8]                     |
| <b>Mandate changes</b>                | Paper mandate variation; add/remove signatories, signing rule | SM&CR; [12]                 |
| <b>Bulk payments / wages</b>          | BACS paper schedule; payee book with per-payee CoP            | CoP [4]                     |
| <b>Pre-approved lending</b>           | Facility request by phone/post; offer and draw-down           | CCA 1974 [13]               |
| <b>International FX payment</b>       | Faxed instruction; rate and fee disclosure                    | FTR [9]                     |
| <b>Dormant reactivation</b>           | Branch reactivation of a dormant account                      | Dormancy handling           |
| <b>Personal / business unlink</b>     | Manual separation of personal data from a relationship        | UK GDPR [15]                |
| <b>Credit ring-fence</b>              | Excludes personal data from business credit decisioning       | UK GDPR Art. 5(1)(c) [15]   |
| <b>Signatory ID register</b>          | Paper KYC/KYB across identity and address evidence            | MLR 2017 [8]; ICO [17]      |
| <b>MTD VAT submission</b>             | Manual VAT return; categorise, review, submit                 | HMRC MTD [14]               |
| <b>Complaint handling</b>             | Paper complaint; intake, triage, decision, FOS route          | FCA DISP [10]               |
| <b>Standing orders &amp; DDs</b>      | Paper mandate / cancellation letter                           | Direct Debit Guarantee [5]  |
| <b>Transaction dispute</b>            | Paper dispute form; chargeback / fraud / DD routes            | PSR 2017 regs 76–77 [6]     |
| <b>Intl. beneficiary onboarding</b>   | Faxed IBAN and wet-ink verification; screening                | MLR 2017 [8], sanctions/PEP |
| <b>Balance certificate</b>            | Written request with postal wait; sealed certificate          | Evidence document           |
| <b>Trusted devices &amp; sessions</b> | Branch visit to reset access; device/session control          | SCA [3], SYSC [12]          |

## SECTION 6

# 6 Regulatory control mapping

The workflows are designed around the controls a production implementation would require. The prototype surfaces each control in the user experience so that a reviewer can locate where an obligation is discharged. The mapping below is indicative of intent, not a statement of certified compliance.

| Framework                     | Obligation reflected                                                           | Ref    |
|-------------------------------|--------------------------------------------------------------------------------|--------|
| <b>Consumer Duty</b>          | Demonstrable good outcomes; fair treatment designed into journeys              | [1]    |
| <b>PSD2 / SCA-RTS</b>         | Strong Customer Authentication and risk-based step-up before sensitive actions | [2][3] |
| <b>Confirmation of Payee</b>  | Account-name verification before a first payment to a new payee                | [4]    |
| <b>Direct Debit Guarantee</b> | Immediate, guaranteed refund route for Direct Debit disputes                   | [5]    |
| <b>PSR 2017 (regs 76–77)</b>  | Refund of unauthorised transactions; correction of errors                      | [6]    |
| <b>APP reimbursement</b>      | Cancel window and reimbursement framing on outbound payments                   | [7]    |
| <b>MLR 2017 / FTR</b>         | Sanctions, PEP and adverse-media screening; payer/payee information            | [8][9] |
| <b>FCA DISP</b>               | Complaint intake, decision, and escalation to the Financial Ombudsman          | [10]   |
| <b>FCA SYSC 9</b>             | Material actions recorded to an audit trail                                    | [12]   |
| <b>HMRC MTD</b>               | Digital preparation and submission of VAT returns                              | [14]   |
| <b>UK GDPR</b>                | Data separation, ring-fencing and purpose limitation                           | [15]   |

The data-protection references reflect the UK regime as reshaped by the Data (Use and Access) Act 2025, whose principal reforms took effect on 5 February 2026; a data-subject complaints-handling duty follows on 19 June 2026 [15]. A production build would need to track that transition — including the regulator's reconstitution as the Information Commission — as configurable policy rather than fixed assumptions.

## SECTION 7

# 7 Engineering and quality approach

Although a concept artefact, the prototype is produced through a disciplined, repeatable pipeline. The pipeline exists because the failure modes of a large single-file front end are predictable, and because a credible artefact must actually work when demonstrated — not merely compile. Four stages are kept distinct:

- **Scaffold** — new workflows are generated from a template that encodes the file conventions and forecloses the recurring defects (state reset, dual-shell wiring, input handling).
- **Review** — a parallel, multi-lens review examines each change for architecture, security, engineering, design-system and state-integrity concerns before it is accepted.
- **Gate** — an automated pre-commit check blocks accessibility, colour-token, markup and build regressions from ever leaving the working tree.
- **Runtime verification** — the built application is driven headlessly through real journeys, and audited against WCAG 2.1 AA [16] on each run, so that behaviour is proven rather than assumed.

### WHY THIS MATTERS

Static review confirms that code exists and is well-formed; only driving the running application confirms that a control actually fires. The pipeline is designed so that the artefact demonstrated to stakeholders is the artefact that has been verified.

## Design system

A single visual language governs spacing, colour, typography and interaction: a fixed spacing scale, the Santander brand red reserved for the primary action on any screen, a warm neutral palette, tabular figures for every monetary value, and exactly one primary call-to-action per view. Consistency here is not decoration; it is what makes seventeen distinct journeys feel like one coherent product.

### SECTION 8

## 8 Accessibility

Accessibility is treated as a build gate rather than a closing audit. Interactive controls carry visible focus states and accessible names; status is never signalled by colour alone; and the interface is measured against WCAG 2.1 AA [16] using an automated engine on every verification run. Where the automated audit identified shortfalls — colour-contrast on small text and unnamed icon-only controls — they were corrected and re-measured to zero on the core screens. Full conformance, including assistive-technology testing, is identified as a pre-production activity.

### SECTION 9

## 9 Limitations and assumptions

The prototype should be read with its boundaries in mind, so that its evidence is neither over- nor under-weighted:

- It is front-end only: no data is stored, transmitted or processed, and no control is enforced server-side.
- All names, figures, accounts and references are illustrative and fictional.
- Regulatory behaviours are represented to communicate intent, not certified for production use.
- It has not been security-tested, load-tested, or assured to production standards.

### SECTION 10

## 10 Indicative next steps

Should the proposition be endorsed, a proportionate path from concept to pilot would be:

- Prioritise the highest-value journeys — by volume, cost-to-serve and customer friction — for a backend-integrated pilot.
- Validate the shortlisted journeys with business customers through structured research, using the prototype as the discussion artefact.
- Complete a full WCAG 2.1 AA conformance review [16], including assistive-technology testing.
- Define the integration, security, resilience and data-protection requirements for a production build, and the target operating model to support it.

## REFERENCES

## References

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- [1] FCA, A New Consumer Duty — Policy Statement PS22/9 (July 2022); FCA Principle 12 and PRIN 2A.
  - [2] The Payment Services Regulations 2017 (SI 2017/752), transposing Directive (EU) 2015/2366 (PSD2); main provisions in force from 13 January 2018 and the secure-communication / SCA provisions from 14 September 2019. Now within the repeal-and-replace programme under the Financial Services and Markets Act 2023.
  - [3] Commission Delegated Regulation (EU) 2018/389 — Regulatory Technical Standards for Strong Customer Authentication and common and secure open standards of communication (SCA-RTS).
  - [4] Pay.UK — Confirmation of Payee (CoP) scheme rules.
  - [5] Bacs — the Direct Debit Guarantee.
  - [6] The Payment Services Regulations 2017, regs 76–77 (unauthorised and incorrectly executed transactions).
  - [7] PSR — Faster Payments APP scams mandatory reimbursement requirement (in force 7 October 2024).
  - [8] The Money Laundering, Terrorist Financing and Transfer of Funds (Information on the Payer) Regulations 2017 (SI 2017/692).
  - [9] Regulation (EU) 2015/847 on information accompanying transfers of funds (the Funds Transfer Regulation), as retained in UK law.
  - [10] FCA Handbook — Dispute Resolution: Complaints (DISP); the Financial Ombudsman Service.
  - [11] FCA Handbook — Banking: Conduct of Business Sourcebook (BCOBS).
  - [12] FCA Handbook — Senior Management Arrangements, Systems and Controls (SYSC 9 record-keeping); the Senior Managers and Certification Regime (SM&CR;).
  - [13] Consumer Credit Act 1974.
  - [14] HMRC — Making Tax Digital for VAT.
  - [15] UK GDPR and the Data Protection Act 2018, as amended by the Data (Use and Access) Act 2025 (Royal Assent 19 June 2025; principal data-protection reforms in Part 5 in force from 5 February 2026; a data-subject complaints-handling duty under s.103 from 19 June 2026; the regulator is being reconstituted as the Information Commission). Article 5(1)(c) — data minimisation and purpose limitation.
  - [16] W3C — Web Content Accessibility Guidelines (WCAG) 2.1, Level AA.
  - [17] ICO — guidance on the pre-employment vetting of candidates: necessity, proportionality and data minimisation when verifying individuals (relevant to the signatory ID register).

*Concept prototype — front-end only. No backend, no live system connections, and no real customer, account or payment data. Prepared for internal discussion.*